



BAULKHAM HILLS HIGH SCHOOL

Assessment Task 2 2014
YEAR 11

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 1 hour
- Write using black or blue pen
- Board-approved calculators may be used
- Show all necessary working in Questions 1-16
- Marks may be deducted for careless or badly arranged work

Total marks – 48

This paper consists of 16 Questions.

1.	A and B are the points $(-5, 12)$ and $(4, 9)$ respectively. Find the coordinates of P which divides AB externally in the ratio 5:2.	2
2.	Evaluate $\sin 75^\circ$ in exact form.	2
3.	Find the acute angle between the lines $y = 4x + 5$ and $6x + 3y = 7$.	2
4.	Solve $3 \cos^2 2\theta = \sin^2 2\theta$, for $0^\circ \leq \theta \leq 180^\circ$.	3
5.	Express $1 - 2 \sin^2 2x$ in terms of $4x$.	1
6.	Show that $\frac{1+\cos x}{\sin x} = \frac{1}{t}$ where $t = \tan \frac{x}{2}$.	2
7.	(i) Show that the points $A\left(\frac{1}{2}, 4\right)$, $P(-3, -3)$ and $Q(1, 5)$ are collinear. (ii) Find the ratio in which A divides the line segment PQ .	2 2
8.	(i) Express $3 \cos x + 4 \sin x$ in the form $A \cos(x - \alpha)$ where $A > 0$ and α is acute. (ii) Hence solve $3 \cos x + 4 \sin x = -3$ for $0^\circ \leq x \leq 360^\circ$	2 2
9.	Prove the identity $\frac{\cot A - \tan A}{\cot 2A} \equiv 2$	3
10.	Solve $8 \sin^2 \theta - 2 \cos^2 \theta = \cos 2\theta$ for $0^\circ \leq \theta \leq 360^\circ$	3

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11.	Use the substitution $t = \tan \theta$ to solve the equation $3\sin 2\theta - 4\cos 2\theta = 4$ for $0^\circ \leq \theta \leq 180^\circ$	4
12.	If $\sin x = \frac{\sqrt{32}}{9}$ and $\sin y = \frac{\sqrt{8}}{3}$ where x and y are acute. Without finding the value of x or y (i) Show that $x = 2y$ (ii) Hence show that the exact value of $\sin 3y = -\frac{10\sqrt{2}}{27}$	2 2
13.	The angle between the line $4x + 3y = 8$ and the line $ax + by + c = 0$ is 45° . Find the possible values of $a : b$.	3
14.	(a) Prove that $\sin(A + B) + \sin(A - B) = 2 \sin A \cos B$ (b) Hence find the exact value of $\sin 157 \frac{1}{2}^\circ \cos 67 \frac{1}{2}^\circ$	1 2
15.	Given that $\sin x + \cos x = a$ and $\cos 2x = b$. Prove that $a^4 - 2a^2 + b^2 = 0$	3
16.	(a) Prove that $\frac{1 - \tan^2 x}{1 + \tan^2 x} = \cos 2x$ (b) Hence prove that if $\cos 2A = \tan^2 B$ then $\cos 2B = \tan^2 A$	2 3

END OF EXAMINATION

Yr 11 EXTENSION 1 TERM 2 2014 SOLUTIONS

1. A(-5, 12) B(4, 9)

$$-5 \div 2$$

$$x = \frac{-10 - 20}{-5 + 2}$$

$$y = \frac{24 - 45}{-3}$$

1 for correct internal division

$$x = \frac{-30}{-3}$$

$$y = \frac{-21}{-3}$$

$$x = 10$$

$$y = 7$$

$\therefore P$ is (10, 7)

2. $\sin 75^\circ = \sin(45^\circ + 30^\circ)$

$$= \sin 45^\circ \cos 30^\circ + \cos 45^\circ \sin 30^\circ$$

$$= \frac{1}{\sqrt{2}} \cdot \frac{\sqrt{3}}{2} + \frac{1}{\sqrt{2}} \cdot \frac{1}{2}$$

$$= \frac{\sqrt{3} + 1}{2\sqrt{2}}$$

$$= \frac{\sqrt{6} + \sqrt{2}}{4}$$

3.

$$m_1 = 4$$

$$m_2 = \frac{-6}{3} = -2$$

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$= \left| \frac{4 - (-2)}{1 - 8} \right|$$

$$= \left| -\frac{6}{7} \right|$$

$$\tan \theta = \frac{6}{7}$$

$$\theta = 40^\circ 36'$$

$$4. \quad 3 \cos^2 2\theta = \sin^2 2\theta \quad 0 \leq \theta \leq 180^\circ$$

$$3 = \tan^2 2\theta \quad 0 \leq 2\theta \leq 360^\circ$$

$$\tan 2\theta = \pm \sqrt{3} \quad \checkmark$$

$$2\theta = 60^\circ, 120^\circ, 240^\circ, 300^\circ$$

$$\theta = \underline{30^\circ, 60^\circ}, \underline{120^\circ, 150^\circ} \quad \checkmark$$

$$5. \quad 1 - 2\sin^2 2x = \cos 4x \quad \checkmark$$

$$6. \quad \text{LHS} = \frac{1 + \cos x}{\sin x}$$

$$= 1 + \frac{1 - t^2}{1 + t^2}$$

$$\frac{2t}{1 + t^2}$$

$$= \frac{1 + t^2 + 1 - t^2}{2t}$$

$$= \frac{2}{2t}$$

$$= \frac{1}{t}$$

$$= \text{RHS as req'd.} \quad \checkmark$$

$$7. \quad i) \quad m_{AP} = \frac{4 - -3}{\frac{1}{2} - -3}$$

$$= \frac{7}{3\frac{1}{2}}$$

$$= 2 \quad \checkmark$$

$$m_{PQ} = \frac{5 - -3}{1 - -3}$$

$$= \frac{8}{4}$$

$$= 2$$

$$\therefore m_{AP}$$

$\therefore A, P, Q$ are collinear $\checkmark$

$$ii) \quad P(-3, -3) \quad Q(1, 5)$$

$$k = 1$$

$$\therefore 4 = \frac{-3 + 5k}{k + 1}$$

$$4k + 4 = -3 + 5k$$

$$7 = k$$

$$\therefore \text{Ratio is } 7:1$$

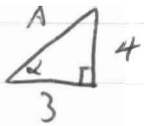
$$8 \text{ (i)} \quad 3 \cos x + 4 \sin x = A \cos(x - \alpha)$$

$$3 \cos x + 4 \sin x = A \cos x \cos \alpha + A \sin x \sin \alpha$$

Equating:

$$\cos x: \quad 3 = A \cos \alpha$$

$$\sin x: \quad 4 = A \sin \alpha$$



$$\sin \alpha = \frac{4}{A}$$



$$\therefore A = 5 \quad \checkmark$$

$$\cos \alpha = \frac{3}{5}$$

$$\alpha = 53^\circ 8' \text{ nearest minute}$$

$$\therefore 3 \cos x + 4 \sin x = 5 \cos(x - 53^\circ 8') \quad \checkmark$$

$$(ii) \quad 3 \cos x + 4 \sin x = -3$$

$$0^\circ \leq x \leq 360^\circ$$

$$5 \cos(x - 53^\circ 8') = -3$$

$$-53^\circ 8' \leq x - 53^\circ 8' \leq 306^\circ 52'$$

$$\cos(x - 53^\circ 8') = -\frac{3}{5}$$

$$x - 53^\circ 8' = 126^\circ 52', 233^\circ 8' \quad \checkmark$$

$$x = 180^\circ, 286^\circ 16' \quad \checkmark$$

$$9. \quad LHS = \frac{\cot A - \tan A}{\cot 2A}$$

$$= \frac{\frac{1}{\tan A} - \tan A}{\frac{1 - \tan^2 A}{2 \tan A}}$$

$$= \frac{1 - \tan^2 A}{\tan A} \times \frac{2 \tan A}{1 - \tan^2 A}$$

$$= 2$$

$$= RHS$$

10.

$$8\sin^2\theta - 2\cos^2\theta = \cos 2\theta$$

$$8\sin^2\theta - 2\cos^2\theta = 2\cos^2\theta - 1 \quad \checkmark$$

$$8\sin^2\theta - 4\cos^2\theta + 1 = 0$$

$$8\sin^2\theta - 4(1 - \sin^2\theta) + 1 = 0$$

$$8\sin^2\theta - 4 + 4\sin^2\theta + 1 = 0$$

$$12\sin^2\theta - 3 = 0$$

$$\sin^2\theta = \frac{3}{12} \quad \checkmark$$

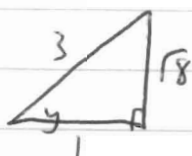
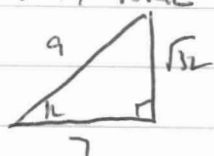
$$\sin^2\theta = \frac{1}{4}$$

$$\sin\theta = \pm \frac{1}{2} \quad \checkmark$$

$$\theta = 30^\circ, 150^\circ, 210^\circ, 330^\circ \quad \checkmark$$

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12



$$i) \sin x = \frac{\sqrt{2}}{9}$$

$$\sin 2y = 2\sin y \cos y \quad \checkmark$$

$$= 2 \cdot \frac{\sqrt{8}}{3} \cdot \frac{1}{3}$$

$$= \frac{2 \times 2\sqrt{2}}{9}$$

$$= \frac{4\sqrt{2}}{9}$$

$$= \frac{\sqrt{2}}{9}$$

$$\text{since } \sin x = \sin 2y \quad (x, y \text{ acute}) \quad \checkmark$$

$$\therefore x = 2y$$

$$ii) \sin 3y = \sin (2y + y)$$

$$= \sin 2y \cos y + \cos 2y \sin y \quad \checkmark$$

$$= \frac{4\sqrt{2}}{9} \cdot \frac{1}{3} + (1 - 2\sin^2 y) \cdot \frac{\sqrt{8}}{3}$$

$$= \frac{4\sqrt{2}}{27} + \frac{\sqrt{8}}{3} - \frac{2\sqrt{8}}{3} \cdot \frac{8}{9}$$

$$= \frac{4\sqrt{2}}{27} + \frac{18\sqrt{2}}{27} - \frac{32\sqrt{2}}{27} \quad \checkmark$$

$$= -\frac{10\sqrt{2}}{27} \text{ as reqd.}$$

11.

$$3 \sin 2\theta - 4 \cos 2\theta = 4$$

$$0 \leq \theta \leq 180^\circ$$

$$\frac{3 \times 2t}{1+t^2} - 4 \frac{(1-t^2)}{1+t^2} = 4 \quad \checkmark$$

$$6t - 4 + 4t^2 = 4 + 4t^2$$

$$6t = 8 \quad \checkmark$$

$$t = \frac{4}{3}$$

$$\tan \theta = \frac{4}{3}$$

$$\theta = 53^\circ 8' \quad \checkmark$$

checking $\theta = 90^\circ$

$$\text{LHS} = 3 \sin 180^\circ - 4 \cos 180^\circ$$

$$= 0 - 4(-1)$$

$$= 4$$

$$= \text{RHS}$$

$\therefore \theta = 90^\circ$ is also a solution $\checkmark$

$$\therefore \theta = 53^\circ 8', 90^\circ \quad \checkmark$$

13.

$$4x + 3y = 8$$

$$y = -\frac{4x}{3} + \frac{8}{3}$$

$$m_1 = -\frac{4}{3}$$

$$ax + by + c = 0$$

$$y = -\frac{ax}{b} - \frac{c}{a}$$

$$m_2 = -\frac{a}{b}$$

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$\tan 45^\circ = \left| \frac{-\frac{4}{3} + \frac{a}{b}}{1 + \frac{4a}{3b}} \right|$$

$$1 = \left| \frac{-4b + 3a}{3b + 4a} \right|$$

$$|4a + 3b| = |3a - 4b|$$

$$4a + 3b = 3a - 4b$$

or

$$4a + 3b = -(3a - 4b)$$

$$a = -7b$$

$$4a + 3b = -3a + 4b$$

$$\therefore \frac{a}{b} = -7$$

$$\therefore a : b = -7 : 1$$

$$7a = b$$

$$\frac{a}{b} = \frac{1}{7}$$

$$\therefore a : b = 1 : 7$$

14.

$$a) \sin(A+B) + \sin(A-B)$$

$$= \sin A \cos B + \cos A \sin B + \sin A \cos B - \sin A \cos B$$

$$= 2 \sin A \cos B$$

$$b) \sin 157.5^\circ \cos 67.5^\circ = \frac{1}{2} [\sin (157.5^\circ + 67.5^\circ) + \sin (157.5^\circ - 67.5^\circ)]$$

$$= \frac{1}{2} (\sin 225^\circ + \sin 90^\circ)$$

$$= \frac{1}{2} (-\sin 45^\circ + 1)$$

$$= \frac{1}{2} \left(1 - \frac{1}{\sqrt{2}} \right)$$

$$= \frac{\sqrt{2} - 1}{2\sqrt{2}}$$

$$= \frac{2 - \sqrt{2}}{4}$$

15.

$$\sin x + \cos x = a$$

$$\cos 2x = b$$

$$\sin^2 x + 2 \sin x \cos x + \cos^2 x = a^2$$

$$\cos^2 2x = b^2$$

$$\sin^2 x + \cos^2 x + \sin 2x = a^2$$

$$1 + \sin 2x = a^2 \quad \checkmark$$

$$(1 + \sin 2x)^2 = a^4$$

$$1 + 2 \sin 2x + \sin^2 2x = a^4$$

$$\text{Now } a^4 - 2a^2 + b^2$$

$$= 1 + 2 \sin 2x + \sin^2 2x - 2(1 + \sin 2x) + \cos^2 2x \quad \checkmark$$

$$= 1 \sin^2 2x + \cos^2 2x - 2 + 2 \sin 2x - 2 \sin 2x \quad \checkmark$$

$$= 0 \quad \text{as req'd.}$$

16.

$$\text{a) LHS} = \frac{1 - \tan^2 x}{1 + \tan^2 x}$$

$$= \frac{1 - \tan^2 x}{\sec^2 x}$$

$$= \frac{1 - \frac{\sin^2 x}{\cos^2 x}}{\frac{1}{\cos^2 x}} \times \cos^2 x$$

$$= \frac{1}{\cos^2 x} \times \cos^2 x$$

$$= \cos^2 x - \sin^2 x$$

$$= \cos 2x$$

$$= \text{RHS as req'd.}$$

$$\text{b) if } \cos 2A = \tan^2 B$$

$$\frac{1 - \tan^2 A}{1 + \tan^2 A} = \tan^2 B$$

$$1 - \tan^2 A = \tan^2 B (1 + \tan^2 A)$$

$$1 - \tan^2 A = \tan^2 B + \tan^2 A \tan^2 B$$

$$1 - \tan^2 B = \tan^2 A + \tan^2 A \tan^2 B$$

$$1 - \tan^2 B = \tan^2 A (1 + \tan^2 B)$$

$$\frac{1 - \tan^2 B}{1 + \tan^2 B} = \tan^2 A$$

$$\text{But from (a) } \frac{1 - \tan^2 B}{1 + \tan^2 B} = \cos 2B$$

$$\therefore \tan^2 A = \cos 2B$$